TENSORING WITH A FIXED MODULE COMMUTES WITH ARBITRARY COLIMITS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-modules/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The description of a colimit of sheaves as the sheafification of the presheaf of colimits of the sections, and the description of the tensor product of sheaves as a sheafification of a presheaf tensor product. Both are invoked at the start of the counted proof (source lines 2637-2644) as a cross-chapter pointer and a pointer to a lemma of this chapter. Reduces the statement to a computation with presheaves of sections.
- The commutation of the tensor product of modules with colimits over a ring, and the description of colimits of modules on a ringed space. Both are invoked inside the counted proof (source lines 2647-2653) as a cross-chapter pointer and a pointer to a lemma of this chapter. Identifies the presheaf of the tensor product with the colimit of the presheaves of the tensored terms, which is the statement.

Source reference. Original lemma lemma-tensor-product-sheafification: sources/stacks-modules/pinned-original-source.tex, source lines 2642–2642. Navigation pointer only: the counted text refers to the Stacks item lemma-tensor-product-sheafification, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original lemma lemma-limits-colimits: sources/stacks-modules/pinned-original-source.tex, source lines 534–534. Navigation pointer only: the counted text refers to the Stacks item lemma-limits-colimits, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own

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Source reference. Original reference sheaves-section-limits-sheaves: sources/stacks-modules/pinned-original-source.tex, source lines 2640–2640. Navigation pointer only: the counted text refers to the Stacks item sheaves-section-limits-sheaves, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference algebra-lemma-tensor-products-commute-with-limits: sources/stacks-modules/pinned-original-source.tex, source lines 2651–2651. Navigation pointer only: the counted text refers to the Stacks item algebra-lemma-tensor-products-commute-with-limits, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space. For any $\mathcal{O}_X$-module $\mathcal{F}$ the functor*

$$Mod(\mathcal{O}_X) \longrightarrow Mod(\mathcal{O}_X), \quad \mathcal{G} \longmapsto \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G}$$

commutes with arbitrary colimits.

Proof. Let I be a preordered set and let $\{\mathcal{G}_i\}$ be a system over I. Set $\mathcal{G} = \operatorname{colim}_i \mathcal{G}_i$. Recall that $\mathcal{G}$ is the sheaf associated to the presheaf $\mathcal{G}' : U \mapsto \operatorname{colim}_i \mathcal{G}_i(U)$, see Sheaves, Section sheaves-section-limits-sheaves. By Lemma lemma-tensor-product-sheafification the tensor product $\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G}$ is the sheafification of the presheaf

$$U \longmapsto \mathcal{F}(U) \otimes_{\mathcal{O}_X(U)} \operatorname{colim}_i \mathcal{G}_i(U) = \operatorname{colim}_i \mathcal{F}(U) \otimes_{\mathcal{O}_X(U)} \mathcal{G}_i(U)$$

where the equality sign is Algebra, Lemma algebra-lemma-tensor-products-commute-with-limits. Hence the lemma follows from the description of colimits in $Mod(\mathcal{O}_X)$, see Lemma lemma-limits-colimits. □